



Viewpoint

Viewpoint | linking long Covid and AD(H)D through neuroimmune dysfunction: A translational framework proposal for precision medicine

Martin Spanoghe^{a,*}, Tomaso Antonacci^a, Nicole Schneider^a, Thomas H.J. Molmans^{b,c}^a Long Covid Belgium, Patient Association, Belgium^b Post-COVID Network Netherlands, Leiden, the Netherlands^c Stichting Long COVID/Dutch Foundation for Long COVID Research, Bosch en Duin, the Netherlands

ARTICLE INFO

Keywords:

Covid-19

SARS-CoV-2

Attention-Deficit/Hyperactivity Disorder

Cognitive impairments

Brain fog

Pathophysiological mechanisms

Kynurenine pathway

Dopaminergic dysfunction

Noradrenergic dysfunction

Immunopsychiatry

1. Introduction

Few medical conditions in recent memory have generated as intense a research focus in merely five years as Long Covid (LC)—a post-acute infection syndrome. Major investigations have explored etiologies, candidate biomarkers and physiological mechanisms such as viral persistence (Proal et al., 2025), yet definitive therapeutic strategy remains elusive. LC's current symptom-based definition reflects this early phase of research: its biomedical complexity, heterogeneity, and systemic nature align poorly with current diagnostic standards, resulting in inconsistent clinical profiles. One manifestation, however, is ubiquitous: profound exhaustion and cognitive dysfunction, widely termed “brain fog”, which in turn can present under the umbrella of post-exertional malaise (PEM), the hallmark of myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS).

Countless LC patients have turned to social media forming large peer communities where lived experiences are shared in real time. A recurring patient-reported observation emerging from these communities is the onset or worsening of attention deficit (hyperactivity) disorder (AD (H)D), or AD(H)D-like symptoms, following COVID-19 infection—despite no prior diagnosis. Both adults and children have

described cognitive, behavioral, and sensory changes consistent with AD (H)D traits. This mirrors observations in ME/CFS and among neurodivergent youth with high levels of disabling fatigue (Sáez-Francàs et al., 2012; Quadt et al., 2024). Common patient-reported features include brain fog,¹ mental exhaustion, impaired attention/concentration, sleep disturbances, impatience, hypersensitivity, conversational difficulties, feeling “different” and, secondarily, feelings of anxiety and depression.

Increasingly recognized by LC healthcare providers, these overlaps suggest a bidirectional relationship between LC and AD(H)D, both involving neuroinflammation and shared neurocognitive symptoms (Braga et al., 2023). Although case series and epidemiological data report AD(H)D as a risk factor for LC and note its higher incidence among LC sufferers (Merzon et al., 2022; Ferrara et al., 2023; Gimbach et al., 2024)—both conditions share hypermobility spectrum disorders as a risk factor—, not all studies support this association (Zemer et al., 2024). This inconsistency underscores the need for deeper investigation into shared phenomenology, symptomatology, prevalence, and potential overlapping pathomechanisms.

* Corresponding author at: Heph-Condorcet, Rue Paul Pastur 11, 7800 Ath, Belgium.

E-mail address: martin.spanoghe@condorcet.be (M. Spanoghe).

¹ Brain fog is a descriptive concept and a heterogeneous condition. The definition used in the context of AD(H)D may differ from other interpretations.

2. Investigating common pathophysiological pathways

Emerging evidence suggests meaningful pathophysiological overlaps between LC and AD(H)D, particularly regarding executive dysfunction. The most recent literature suggests four major converging mechanisms:

1) Dysfunction in frontal–striatal–hippocampal systems.

Neuroimaging and pathophysiological studies feature dysfunction in the prefrontal cortex, basal ganglia, striatum, and hippocampus—regions responsible for executive control and cognitive modulation. These deficits involve disrupted catecholaminergic, glutamatergic and GABAergic neurotransmission, alongside signs of neuron senescence (da Silva et al., 2023; Mancini et al., 2023; Yang et al., 2024; Fujimoto et al., 2025).

2) Neuroimmune dysregulation of catecholamine systems.

Neuroimmune dysregulation of dopamine and noradrenaline pathways leads to altered immune modulation involving changes in T-cells and macrophage/microglia activity, cytokine imbalances, and disrupted neuroimmune communication via the autonomic nervous system (Sarkar et al., 2010; Feng and Lu, 2021). Hypodopaminergia in AD(H)D may also reduce immune competence, potentially increasing vulnerability to post-acute viral syndromes like LC.

3) Dysregulated tryptophan-kynurenine metabolism.

Overactivation of the kynurenine pathway produces neurotoxic and pro-inflammatory metabolites disrupting glutamatergic—and indirectly catecholaminergic—transmission (Rassoulpour et al., 2005; Cysique et al., 2023; Cavaleri et al., 2024). Serotonin, synthesized from tryptophan, not only modulates noradrenergic and dopaminergic neurotransmission but is itself regulated reciprocally by these monoaminergic systems, thereby establishing dynamic feedback loops (Guiard et al., 2008).

4) Mitochondrial bioenergetic defects.

Mitochondrial dysfunction underlies the (neuro)pathogenesis of ME/CFS and LC, driving fatigue, PEM, and cognitive deficits, likely through oxidative stress, increased reactive oxygen species (Molnar et al., 2024; Shankar et al., 2025), and suppression of mitochondrial complex gene expression (Xu et al., 2025). Comparable bioenergetic impairments occur in AD(H)D models (Verma et al., 2016). The tryptophan-kynurenine pathway is associated with mitochondrial dysfunction (Tanaka et al., 2022).

Clinical observations also report partial symptomatic improvement in LC patients treated off-label with medications used in AD(H)D, including methylphenidate, guanfacine, low-dose lithium, and (lis) dexamfetamine (Krishnan et al., 2022; Koonce and Martin, 2024; Fesharaki-Zadeh et al., 2023), although side effects, pre-existing autonomic dysregulation, and PEM remain limiting factors. These observations suggest that a subset of LC patients may experience a distinct form of dopaminergic/noradrenergic dysfunction—analogueous to a neuroimmune-mediated ‘depression’—potentially triggered by infection and perpetuated by viral persistence (Coleon et al., 2025).

3. Rethinking therapeutics and precision medicine through immune subtyping

There is growing recognition in psychiatry that traditional neurotransmitter-based treatments fail to address the complexity of many neuropsychiatric disorders. Regional brain specificity, circuit-level targeting and immune modulation remain underdeveloped. New approaches focusing on neuroinflammation and immune dysregulation are increasingly relevant.

Given their multifaceted phenotypes, LC, and to a lesser extent AD (H)D, may benefit from precision strategies that use immune-based subtyping. This approach, already advancing in depression and psychosis research (Lynall et al., 2020), could enhance diagnostic resolution and enable more targeted mechanism-driven interventions. Immune-based stratification could guide selective personalized interventions, such as targeted immunomodulation or tailored stimulant therapy,

focused on neurocognitive and immunological endophenotypes rather than broad syndromic classifications.

4. Toward a translational research framework to restore neuroimmune homeostasis

Therapeutic innovation in post-acute viral conditions such as LC should prioritize restoring disrupted neuroimmune and metabolic homeostasis. Modulating disrupted neurotransmitter and immunometabolic pathways may help recalibrate both endogenous immune and neural systems, could support recovery and might even mitigate the possible clinical effects of surinfection, all without the need for long-term psychopharmacological stimulation. We therefore propose a translational research model grounded in convergent LC–AD(H)D mechanisms. Key priorities include:

- **Phenotypic Stratification:** Identifying LC patients with significant cognitive/executive symptoms for biomarker profiling (e.g., neuroimaging, immune–metabolic components, omics data).
- **Clinical Trials Designing:** Evaluating the efficacy and safety of AD (H)D-targeted pharmacotherapies in LC cohorts selected for shared biomarkers.
- **Mechanism-Driven Interventions:** Exploring novel interventions that address the root causes of neuroinflammation and immune-metabolic dysfunction simultaneously, rather than merely compensating for neurotransmitter deficits.

5. Conclusion

A translational framework linking LC and AD(H)D may reveal shared neuroimmune pathomechanisms of executive dysfunction, involving neuroinflammation, mitochondrial dysfunction, and immunometabolism imbalances. Collaboration between psychiatry, immunology, infectiology and patient communities is essential to effectively translating these findings into clinical practice. By targeting underlying mechanisms rather than symptomatic relief, precision immunopsychiatry offers a promising avenue to restore neuroimmune homeostasis and improve outcomes in LC and other neuroinflammatory conditions.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT with caution only to improve readability. After using this tool, the authors reviewed and edited the content as needed and took full responsibility for the content of the publication.

CRediT authorship contribution statement

Martin Spanoghe: Writing – original draft, Investigation, Conceptualization. **Tomaso Antonacci:** Writing – review & editing, Investigation, Conceptualization. **Nicole Schneider:** Writing – review & editing. **Thomas H.J. Molmans:** Writing – review & editing, Supervision.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or non-profit sectors.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

References

- Braga, J., Lepra, M., Kish, S.J., Rusjan, P.M., Nasser, Z., Verhoeff, N., Vasdev, N., Bagby, M., Boileau, I., Husain, M.I., Kolla, N., Garcia, A., Chao, T., Mizrahi, R., Faiz, K., Vieira, E.L., Meyer, J.H., 2023. Neuroinflammation after COVID-19 with persistent depressive and cognitive symptoms. *JAMA Psychiat.* 80 (8), 787–795. <https://doi.org/10.1001/jamapsychiatry.2023.1321>.
- Cavaleri, D., Crocamo, C., Morello, P., Bartoli, F., Carrà, G., 2024. The Kynurenine Pathway in Attention-Deficit/Hyperactivity Disorder: a Systematic Review and Meta-Analysis of Blood Concentrations of Tryptophan and its Catabolites. *J. Clin. Med.* 13 (2), 583. <https://doi.org/10.3390/jcm13020583>.
- Coleon, A., Larrous, F., Kergoat, L., Tichit, M., Hardy, D., Obadia, T., Kornobis, E., Bourhy, H., Dias de Melo, G., 2025. Hamsters with long COVID present distinct transcriptomic profiles associated with neurodegenerative processes in brainstem. *Nat. Commun.* 16, 6714. <https://doi.org/10.1038/s41467-025-62048-7>.
- Cysique, L.A., Jakabek, D., Bracken, S.G., Allen-Davidian, Y., Heng, B., Chow, S., Dehghani, M., Staats Pires, A., Darley, D.R., Byrne, A., Phetsouphanh, C., Kelleher, A., Dore, G.J., Matthews, G.V., Guillemin, G.J., Brew, B.J., 2023. The kynurenine pathway relates to post-acute COVID-19 objective cognitive impairment and PASC. *Ann. Clin. Transl. Neurol.* 10 (8), 1338–1352. <https://doi.org/10.1002/acn3.51825>.
- da Silva, B.S., Grevet, E.H., Silva, L.C.F., Ramos, J.K.N., Rovaris, D.L., Bau, C.H.D., 2023. An overview on neurobiology and therapeutics of attention-deficit/hyperactivity disorder. *Discov. Mental Health* 3 (1), 2. <https://doi.org/10.1007/s44192-022-00030-1>.
- Feng, Y., Lu, Y., 2021. Immunomodulatory Effects of Dopamine in Inflammatory Diseases. *Front. Immunol.* 12, 663102. <https://doi.org/10.3389/fimmu.2021.663102>.
- Fesharaki-Zadeh, A., Lowe, N., Arnsten, A.F., 2023. Clinical experience with the α 2A-adrenoceptor agonist, guanfacine, and N-acetylcysteine for the treatment of cognitive deficits in “Long-COVID19”. *Neuroimmunology Reports* 3, 100154. <https://doi.org/10.1016/j.jnerep.2022.100154>.
- Ferrara, F., Zavaleta, E., Vitiello, A., Villalobos, J.A., Zovi, A., Langella, R., Serrano, B., Trama, U., Arguedas, S., Nava, E., Diaz, J.P., Bianco, E., Russo, G., 2023. The direct correlation between Covid-19 and attention Deficit/Hyperactivity Disorder (ADHD). *Eur. Neuropsychopharmacol.* 73, 62–64. <https://doi.org/10.1016/j.euroneuro.2023.04.010>.
- Fujimoto, Y., Abe, H., Eiro, T., Tsugawa, S., Tanaka, M., Hatano, M., Nakajima, W., Ichijo, S., Arisawa, T., Takada, Y., Kimura, K., Sano, A., Hirahata, K., Sasaki, N., Kimura, Y., Takahashi, T., 2025. Systemic increase of AMPA receptors associated with cognitive impairment of long COVID. *Brain. Communications* 7 (5), cfaf337. <https://doi.org/10.1093/braincomms/cfaf337>.
- Gimbach, S., Vogel, D., Fried, R., Faraone, S.V., Banaschewski, T., Buitelaar, J., Döpfner, M., Ammer, R., 2024. ADHD medicine consumption in Europe after COVID-19: catch-up or trend change? *BMC Psychiatry* 24, 112. <https://doi.org/10.1186/s12888-024-05505-9>.
- Guiard, B.P., El Mansari, M., Merali, Z., Blier, P., 2008. Functional interactions between dopamine, serotonin and norepinephrine neurons: an in-vivo electrophysiological study in rats with monoaminergic lesions. *Int. J. Neuropsychopharmacol.* 11 (5), 625–639. <https://doi.org/10.1017/S1461145707008383>.
- Koonce, R.M., Martin, B.C., 2024. The association between suspected long-COVID and stimulant prescribing in the United States. *Psychiatry Res.* 331, 115686. <https://doi.org/10.1016/j.psychres.2023.115686>.
- Krishnan, K., Lin, Y., Prewitt, K.R.M., Potter, D.A., 2022. Multidisciplinary approach to brain fog and related persisting symptoms post COVID-19. *J. Health Serv. Psychol.* 48 (1), 31–38. <https://doi.org/10.1007/s42843-022-00056-7>.
- Lynall, M.E., Turner, L., Bhatti, J., Cavanagh, J., de Boer, P., Mondelli, V., Jones, D., Drevets, W.C., Cowen, P., Harrison, N.A., Pariante, C.M., Pointon, L., Clatworthy, M. R., Bullmore, E., Neuroimmunology of Mood Disorders and Alzheimer’s Disease (NIMA) Consortium, 2020. Peripheral Blood Cell-Stratified Subgroups of Inflamed Depression. *Biol. Psychiatry* 88 (2), 185–196. <https://doi.org/10.1016/j.biopsych.2019.11.017>.
- Mancini, M., Natoli, S., Gardoni, F., Di Luca, M., Pisani, A., 2023. Dopamine transmission imbalance in neuroinflammation: perspectives on long-term COVID-19. *Int. J. Mol. Sci.* 24 (6), 5618. <https://doi.org/10.3390/ijms24065618>.
- Merzon, E., Weiss, M., Krone, B., Cohen, S., Ilani, G., Vinker, S., Cohen-Golan, A., Green, I., Israel, A., Schneider, T., Ashkenazi, S., Weizman, A., Manor, I., 2022. Clinical and Socio-Demographic Variables Associated with the Diagnosis of Long COVID Syndrome in Youth: a Population-based Study. *Int. J. Environ. Res. Public Health* 19 (10), 5993. <https://doi.org/10.3390/ijerph19105993>.
- Molnar, T., Lehoczi, A., Fekete, M., Varnai, R., Zavari, L., Erdo-Bonyar, S., Simon, D., Berki, T., Csécséi, P., Ezer, E., 2024. Mitochondrial dysfunction in long COVID: mechanisms, consequences, and potential therapeutic approaches. *GeroScience* 46 (5), 5267–5286. <https://doi.org/10.1007/s11357-024-01165-5>.
- Proal, A.D., Aleman, S., Bomsel, M., Brodin, P., Buggert, M., Cherry, S., Chertow, D.S., Davies, H.E., Dupont, C.L., Deeks, S.G., Ely, E.W., Fasano, A., Freire, M., Geng, L.N., Griffin, D.E., Henrich, T.J., Hewitt, S.M., Iwasaki, A., Krumholz, H.M., Locci, M., Peluso, M.J., 2025. Targeting the SARS-CoV-2 reservoir in long COVID. *Lancet Infect. Dis.* 25 (5), e294–e306. [https://doi.org/10.1016/S1473-3099\(24\)00769-2](https://doi.org/10.1016/S1473-3099(24)00769-2).
- Quadt, L., Csécséi, J., Bond, R., Harrison, N.A., Critchley, H.D., Davies, K.A., Eccles, E., 2024. Childhood neurodivergent traits, inflammation and chronic disabling fatigue in adolescence: a longitudinal case–control study. *BMJ Open* 14, e084203. <https://doi.org/10.1136/bmjopen-2024-084203>.
- Rassoulpour, A., Wu, H.-Q., Ferré, S., Schwarcz, R., 2005. Nanomolar concentrations of kynurenic acid reduce extracellular dopamine levels in the striatum. *J. Neurochem.* 93, 762–765. <https://doi.org/10.1111/j.1471-4159.2005.03134.x>.
- Sáez-Francás, N., Alegre, J., Calvo, N., Antonio Ramos-Quiroga, J., Ruiz, E., Hernández-Vara, J., Casas, M., 2012. Attention-deficit hyperactivity disorder in chronic fatigue syndrome patients. *Psychiatry Res.* 200 (2–3), 748–753. <https://doi.org/10.1016/j.psychres.2012.04.041>.
- Sarkar, C., Basu, B., Chakroborty, D., Dasgupta, P.S., Basu, S., 2010. The immunoregulatory role of dopamine: an update. *Brain Behav. Immun.* 24 (4), 525–528. <https://doi.org/10.1016/j.bbi.2009.10.015>.
- Shankar, V., Wilhelm, J., Curtis, E.J., Michael, B., Cervantes, L., Mallajosyula, V., Davis, R.W., Snyder, M., Younis, S., Robinson, W.H., Shankar, S., Mischel, P.S., Bonilla, H., Davis, M.M., 2025. Oxidative stress is a shared characteristic of ME/CFS and Long COVID. *PNAS* 122 (28), e2426564122. <https://doi.org/10.1073/pnas.2426564122>.
- Tanaka, M., Szabo, A., Spekter, E., Polyák, H., Tóth, F., Vécsei, L., 2022. Mitochondrial Impairment: a Common motif in Neuropsychiatric Presentation? the link to the Tryptophan–Kynurenine Metabolic System. *Cells* 11 (16), 2607. <https://doi.org/10.3390/cells11162607>.
- Verma, P., Singh, A., Nthenge-Ngumbau, D.N., Rajamma, U., Sinha, S., Mukhopadhyay, K., Mohanakumar, K.P., 2016. Attention deficit-hyperactivity disorder suffers from mitochondrial dysfunction. *BBA Clin.* 6, 153–158. <https://doi.org/10.1016/j.bbacli.2016.10.003>.
- Xu, W.T., An, X.B., Chen, M.J., Ma, J., Wang, X.Q., Yang, J.N., Wang, Q., Wang, D.Y., Wu, Y., Zeng, L., Qu, Y., Zhao, B., Ai, J., 2025. A Gene Cluster of Mitochondrial Complexes Contributes to the Cognitive Decline of COVID-19 Infection. *Mol. Neurobiol.* 62 (6), 6869–6883. <https://doi.org/10.1007/s12035-024-04471-3>.
- Yang, L., Kim, T.W., Han, Y., Nair, M.S., Harschnitz, O., Zhu, J., Chen, S., 2024. SARS-CoV-2 infection causes dopaminergic neuron senescence. *Cell Stem Cell* 31 (2), 196–211. <https://doi.org/10.1016/j.stem.2023.12.012>.
- Zemer, V.S., Manor, I., Weizman, A., Cohen, H.A., Hoshen, M., Caspi, N.M., Cohen, S., Faraone, S.V., Shahar, N., 2024. The influence of COVID-19 on attention-deficit/hyperactivity disorder diagnosis and treatment rates across age, gender, and socioeconomic status: a 20-year national cohort study. *Psychiatry Res.* 339, 116077. <https://doi.org/10.1016/j.psychres.2024.116077>.